Module 6 Tangent: Macros

Macros

Code is data, data is code.

0.

Assignments

So little to do and so much time… wait.

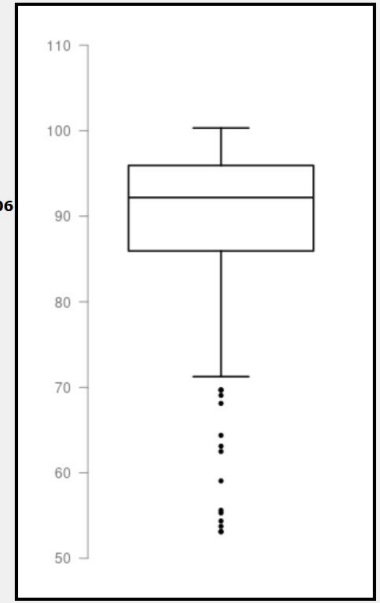
Evaluator Final

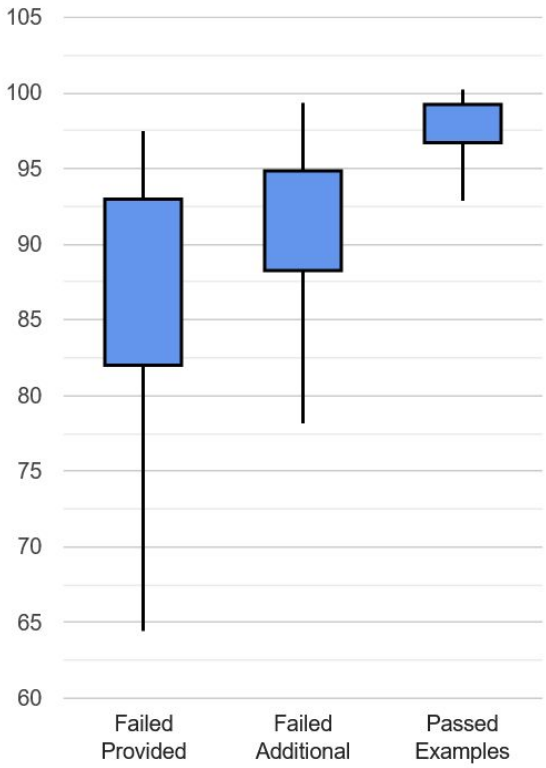
Average 89.8%, Median 92.2%

- Overall good; hardest project part!
- Review for issues & double jeopardy.
- Many compilation failures - check comments for requested changes.
- Opted to remove consideration of EvaluateException.ast from grading.
- Most missed test cases:
  - Static Scoping (?)
  - Evaluation Order, especially errors
  - ObjectExpr, especially Methods

Population size: 370
Median: 92.19
Minimum: 53.12
Maximum: 100.31
First quartile: 85.94
Third quartile: 95.94
Interquartile Range: 10
Outliers: 53.12 53.12 53.75 54.37 55.31 55.62 59.06 62.50 63.12 64.38 68.12 69.06 69.69 69.69

Mean
Geometric mean
Harmonic mean
Median
Mode
Standard Deviation
Variance
Mean absolute deviation
Range
Interquartile range
Quartiles
All dispersion data





Partitioned Evaluator Grades

Overall (excl. Exercise 4)

Average 91.4%, Median 93.6%

- Overall strong, left-skewed.
- Most extreme cases involve missing submissions - submit your tests, and definitely submit your assignments.
- Finale is 33% - make sure to go back and fix previous issues!
- If you're struggling with concepts, go to office hours and *ask questions*.

Population size: 376
Median: 93.595
Minimum: 55.07
Maximum: 99.26
First quartile: 89.8825
Third quartile: 95.9
Interquartile Range: 6.0175
Outliers: 55.07 59.96 63.38 66.45 67.03 67.80 67.86 68.19 69.66 71.10 72.57 72.85 73.52 73.62 73.87 73.89 74.80 74.96 75.33 75.41 75.99 76.94 77.34 77.58 77.81 79.00 79.16 79.65 80.09 80.80

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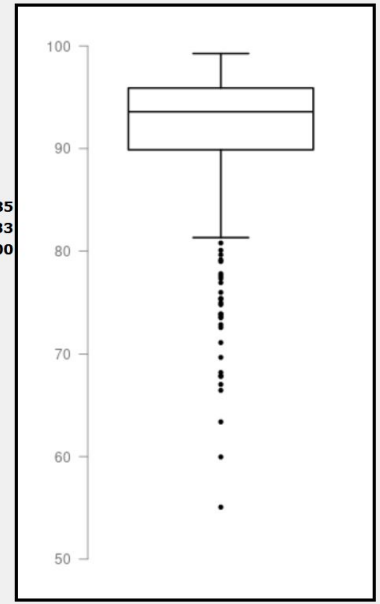

1.

Macros

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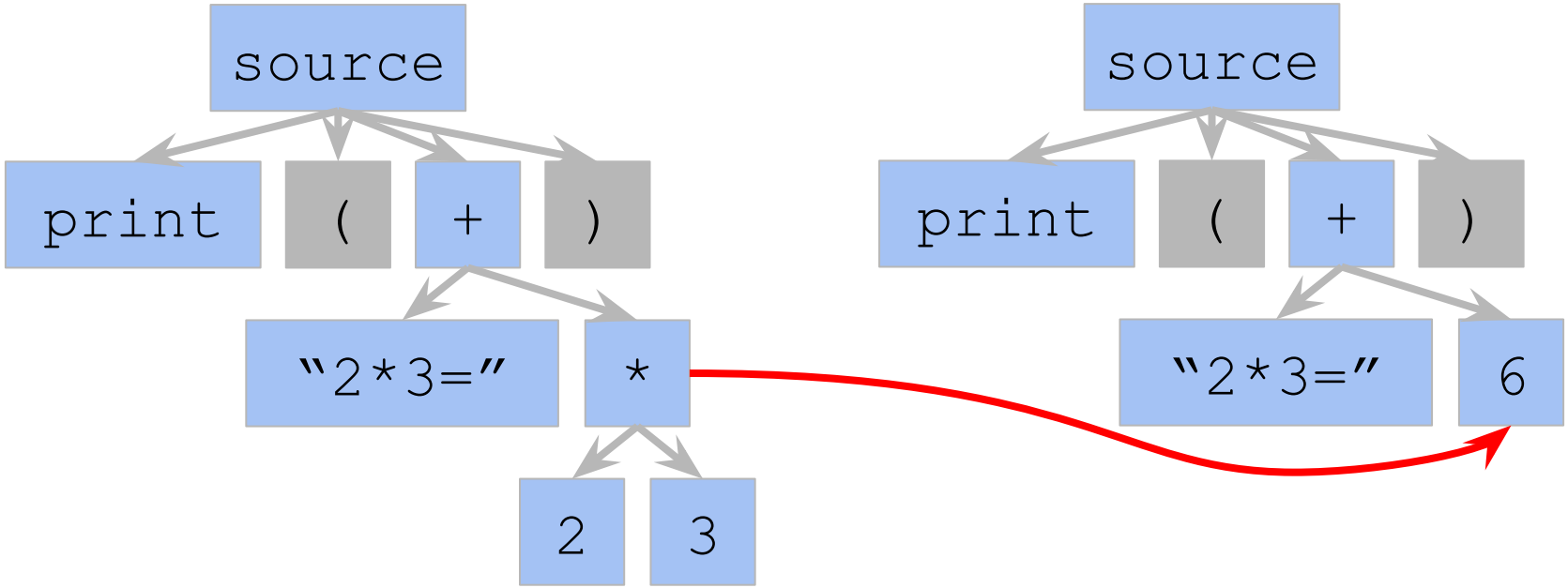


Code is Data, Data is Code

Macros

Definition

A macro is a function that takes code as input and returns code as output.

- Instead of evaluating code, it is transformed into new output code.
  - May also evaluate at compile-time.
- Lisp languages popularized the idea of “code is data, data is code”.
  - Racket is a language to look at.

class-c-new.rkt (define ...)

```
#lang racket/base
(require "class-internal.rkt"
         "class-c-old.rkt"
         "class-wrapped.rkt"
         "../contract/base.rkt"
         "../contract/combinator.rkt"
         (for-syntax racket/base
                     syntax/name
                     syntax/stx))

(provide class/c2)

(define-syntax (class/c2 stx)
  (define-values (opaque? args)
    (syntax-case stx ()
      [(_ #:opaque args ...)
       (values #t (syntax->list #'(args ...)))]
      [(_ args ...)
       (let ()
         (define stx-args (syntax->list #'(args ...)))
```

```c
#define SQR(n) n * n

int x = 1;
int sqr = SQR(x + 2);
printf("%d\n", sqr); //5..?

int sqr = SQR(x + 2);
        = n * n;
        = x + 2 * x + 2;
        = 1 + 2 * 1 + 2;
        = 5;
```

```c
#define SQR(n) (n) * (n)

int x = 1;
int sqr = SQR(x+2);
printf("%d\n", sqr); //9!

int sqr = SQR(x + 2);
        = (n) * (n);
        = (x + 2) * (x + 2);
        = (1 + 2) * (1 + 2);
        = 9;
```

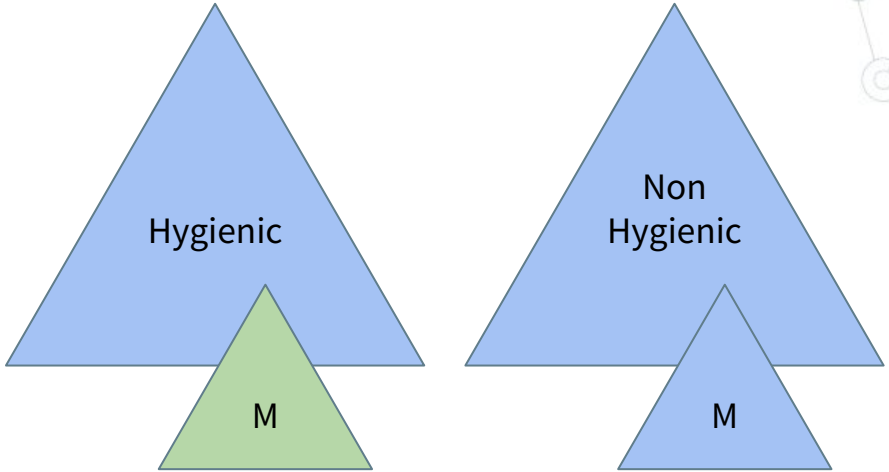
Even C has Macros… Kinda

Hygiene

Definition

Hygiene refers to the scope used to resolve identifiers in macros.

- A *hygienic* macro resolves identifiers at the definition site.
- A *non-hygienic* macro resolves identifiers at the calling site.
  - Similar to dynamic scoping.
- There's a time and place for both, but hygienic macros are safer.



```
//hygienic macro

int x = 1;
macro M() {
  return x;
}

function f() {
  int x = 2;
  return M();
}

f(); //1
```

```
//non-hygienic macro

int x = 1;
macro M() {
  return x;
}

function f() {
  int x = 2;
  return M();
}

f(); //2
```

Macro Hygiene

Module 6: Compilation

Compilation

Time for a Module 6 crash course in one lecture.

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